Diagnostic features:

- It is a green, flattened stem.
- Phylloclade (Cladophyll) is the stem modification, perform the function of leaves.
- Leaves are modified into spines for xerophytic adaptation.

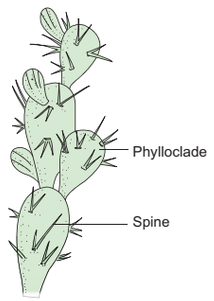


Figure 13: Phylloclade - *Opuntia*

Execise: 14

Special inflorescence- Cyathium

Aim: To study and identify the special type of inflorescence

Principle: Group of flowers arising from a branched or unbranched axis with a definite pattern. The inflorescences do not show any of the development pattern types are classified under special type of inflorescence. Function of inflorescence is to display the flowers for effective pollination and facilitate seed dispersal.

Requirements : Fresh specimen of cyathium inflorescence.

Diagnostic features:

- Special type of inflorescence consists of small unisexual flowers.
- Centrally located single female flower surrounded by male flowers.
- Male flower represented by only stamen and female flower represented only by pistil.
- Involucre protect flowers and consist of nectar.

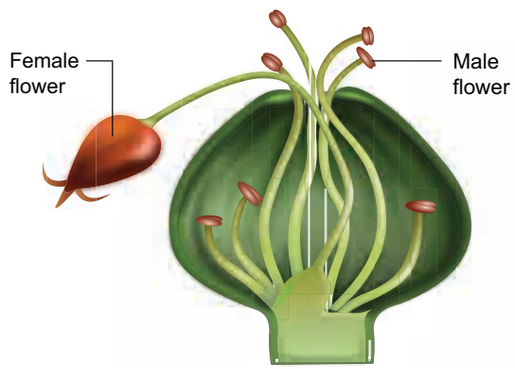


Figure 14: Cyathium inflorescence

Exercise: 15

Aggregate fruit - *Polyalthia*

Aim : To study and identify aggregate fruit type.

Principle: Fruit is a fertilized and ripened ovary. Fruits are classified into various types as, simple fruits, aggregate fruits and multiple fruits. Aggregate fruits develop from a single flower having an apocarpous pistil, each of the free carpel is develops into a simple fruitlet. A collection of simple fruitlets makes an aggregate fruits.

Requirements: Fresh specimen of *Polyalthia* fruit.

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